

ArnoTek Arduino Utilities – Debug

The ArnoTek Debug utility conditionally defines standard Arduino statements to enable generation of debug statements at compile time. These debug statements may be used in conjunction with other standard Arduino statements in the application code if desired.

If enabled, the debugging statements are generated by the compiler as normal standard Arduino statements. If not enabled, the debugging statements are defined as null and they are not generated for compilation. Thus, when the debugging statements are not generated, the statements do not utilize any Arduino program memory and do not consume any CPU cycles at run time.

To use this utility, install the **ArnoTek_Debug.h** file in your Arduino libraries directory. This will make the utility accessible to all of your projects. (ArnoTek utilities and global variables are prefixed with “ArnoTek” in an effort to prevent conflicts with include files from other developers.)

The utility conditionally defines the following statements for use in debugging:

ArnoTek_DebugInit(x);

This statement defines a conditional **Serial.begin** statement and takes one parameter that defines the speed of the serial connection. If standard **Serial.print** statements are being used in addition to the debugging statements, this statement should not be used. The standard **Serial.begin** statement should be used.

ArnoTek_Debug(x);

This statement defines a conditional **Serial.print** statement and takes one parameter that specifies the data to be output. This statement does not generate a new line character.

ArnoTek Arduino Utilities – Debug

ArnoTek_DebugLn(x);

This statement defines a conditional **Serial.println** statement and takes one parameter that specifies the data to be output. This statement generates a new line character.

ArnoTek_Debug2(x, y);

This statement defines a conditional **Serial.print** statement and takes two parameters. The first parameter is the data to be output and the second parameter is the formatting specification. This statement does not generate a new line character.

ArnoTek_DebugLn2(x, y);

This statement defines a conditional **Serial.println** statement and takes two parameters. The first parameter is the data to be output and the second parameter is the formatting specification. This statement generates a new line character.

ArnoTek_DebugFlush();

This statement defines a conditional **Serial.flush** statement.

ArnoTek_DebugDelay(x);

This statement defines a conditional **delay** statement and takes one parameter that specifies the number of milliseconds to delay program execution.

Anytime you wish to use this utility in your code, you must tell the compiler to include the **.h** file in your application. This is accomplished by placing the **#include <ArnoTek_Debug.h>** compiler directive at the beginning of your code. You also need to indicate if the debugging statements are to be generated at compile time by placing a **#define ArnoTek_Debug**

ArnoTek Arduino Utilities – Debug

compiler directive ***BEFORE*** the **#include <ArnoTek_Debug.h>**. To generate the debugging statements, define the ArnoTek_Debug compiler variable and set its value to 1 as follows:

```
#define ArnoTek_Debug = 1
```

To not generate the debugging statements, define the ArnoTek_Debug compiler variable and set its value to 0 as follows:

```
#define ArnoTek_Debug = 0
```

The debugging statements may be used as desired throughout the application code. The conditional statements function just like their standard Arduino statement counterparts with the exception that there are additional debug statements when two parameters are required. For example, when a variable is desired to be output with hex formatting.

The key benefit of using these conditional debugging statements is that they are generated at compile time and if they are not utilized, they do not use any Arduino program memory or consume any CPU cycles.